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Class Assignment nº3

Linux and NS2 installation

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1. Introduction

This assignment aims to help you installing the software that will be used during the next practical lessons of this course.

2 – Description of the software to use

The following software will be used:
• Linux
• Network Simulator 2 (NS2)

2.1 – Linux

To test network commands and communication with sockets, a Linux machine will be used. Any of the existing Linux distributions can be used, as long as it supports the development tools.

2.2 - NS2

NS2 is a discrete computer network simulator that will allow you to test the various protocols studied in this course. It works through a command line and uses the C++ language and Tcl (Tool Command Language) / OTcl. In our lessons we will only use the Tcl/OTcl scripting language. It will be used to create the simulation files. The result of the NS2 simulations is written in trace files where each line represents an event of a data packet, from its sending node to the receiver node. In addition, the NS2 can also generate trace files for the NAM (Network Animator), a visualization tool that allows you to graphically visualize the transmission of data packets between network nodes.

Some supplementary documents on how NS2 works:

- http://www.isi.edu/nsnam/ns/tutorial/index.html
- http://www-sop.inria.fr/members/Eitan.Altman/COURS-NS/n3.pdf

3 – Installation of the software

In this section, 3 alternatives are proposed for installing the software necessary for the chair:

Option 1. Use a virtual machine created specifically for this course and which already has all the necessary software installed (recommended option).

Option 2. Full installation from the scratch (you will need to install Linux, NS2 and NAM) - this option is suitable for machines with low processing speed and/or low RAM memory;

The 2 options are described in more detail below.

3.1 – OPTION 1: Use of a virtual machine with all software pre-installed (**recommended option**)

In order to facilitate the installation work, a Linux virtual machine was created with all the necessary software already installed. This machine is available in OVA format, compatible with VMware and VirtualBox.

To install the virtualization software go to:

VMWare: https://my.vmware.com/web/vmware/downloads
VirtualBox: https://www.virtualbox.org/

Linux virtual machine image:
Ubuntu VM (UC Student)
- File of about 2.6GB
- Ubuntu 18.04 (64-bit)

Credentials to access the system:
login: user
password: password

Important note:
- See the "Problems with Virtual Box" section later in this document in case you experience problems with the VirtualBox machine.

3.2 – OPTION 2: Full installation

In this option you will have to start by installing the Linux Operating System of your choice on your machine and then install NS2.

3.2.1 – Linux installation

You can use one of several available distributions (eg Ubuntu: http://www.ubuntu.com/, Fedora: http://fedoraproject.org/). On each of the distributions' websites you will find information about the installation. You should install the development tools.

3.2.2 – NS2 installation

NS2 can be installed on a machine with a Microsoft Windows operating system, or on any Unix (FreebSD, Linux, etc.).

To obtain the NS2, go to the site http://www.isi.edu/nsnam/ns/ and download the desired version. You will also find instructions for installing on Windows and Linux. It is recommended to install the ns-allinone-2.35 package, which contains all the necessary programs including NAM. There are often

problems during the installation of NS so it is advisable to read some websites with additional installation information:

• Installation of NS2 on Microsoft Windows:
http://ns2ultimate.tumblr.com/post/13334371281/ns-2-35-final-installation-guide

• Installation of NS2 on Ubuntu:
http://mithunme.wordpress.com/2013/02/12/how-to-install-ns-2-in-ubuntu-12-10/

• Installation of NS2 on Fedora:
http://www.nsnam.com/2013/09/installing-network-simulator-2-ns-2-35-in-fedora-19.html

4 – Checking NS2 is installed on Linux

In order to verify the installation of NS2 on Linux you should run the following commands:

```
$ cd {pasta de instalação do NS2}/ns-2.35/tcl/ex
$ ns simple.tcl
```

You should get a small simulation as a response that will be displayed in nam (Network animator)

Note:
In the case of the pre-installed virtual machine, the NS2 installation folder is:

```
~user/ns-allinone-2.35
```

5 – Problems with VirtualBox

A few notes on frequent problems with the VirtualBox virtualization software.

Note 1:
- *Guest Additions* allow for better integration of your host with the virtualized machine. In addition to performance improvements, it also allows clipboard sharing and folder sharing between the host system and the virtualized system. Remove existing *Guest Additions* and install the latest ones suitable for your system:

• Remove existing tools:

```
$ sudo apt remove --purge virtualbox-guest-additions-iso
```

• Then reinstall the correct Guest Additions for your Virtual Box version by selecting the option: "Devices → Insert Guest Additions CD image..."
(https://en.wikibooks.org/wiki/VirtualBox/Setting_up_an_Ubuntu_virtual_machine)

• At the end, reboot the machine.

To activate clipboard and folder sharing you have to activate the respective options in the VirtualBox menu.

Note 2:
The virtual machine is very slow.

Possible solution:
- Make sure that in the machine's BIOS the hardware support option for virtual machines is enabled.
- In the properties of the Ubuntu virtual machine (“Settings”) increase the video memory and change the number of processors used to 2 (if your machine allows it).

Note 3:
Problems with graphics:
- Windows being drawn on the screen slowly

- It is not possible to automatically resize the virtual screen (it is not possible to select the option “View → Auto-resize Guest Display” or it is grayed out)

Possible solution:
- Try removing Guest Additions (see Note 1)

Note 4:
If you have problems with the Ubuntu video resolution in VirtualBox, run the following command in Linux:

```
$ sudo apt install virtualbox-guest-dkms virtualbox-guest-utils virtualbox-guest-x11
```

Note 5:
Can't see shared folders in Virtual Box:

- Make sure you have installed the *Guest Additions*.
- Try accessing */media/sf_name_of_shared_folder* . If you still don't have access, do:

```
$ sudo adduser {o seu username} vboxsf
```

Nota 6:
O ponteiro do rato desaparece dentro da VM.

Por vezes é necessário alterar a configuração da máquina virtual localizada em:
`Machine -> Settings -> System -> Motherboard` *tab*
Escolha a opção **Enable I/O APIC**.

Note 6:
The mouse pointer disappears inside the VM.

Sometimes it is necessary to change the configuration of the virtual machine located at:
`Machine -> Settings -> System -> Motherboard` *tab*
Choose the option Enable I/O APIC